

CLIENT REPORT

Chelsea Housing and Public Health

Help Chelsea identify housing-related public-health risks earlier and prioritize limited inspection capacity.

REPORT REFERENCE

AUD-A57071CBE3

GENERATION DATE

26 August 2026

BÓVEDA VERSION

Alpha 1.0.2

In this report

01	Executive Summary
02	Project Overview
03	Data & Populations
04	Results & Evaluation
05	Feature / Driver Evidence
06	What Bóveda Found
07	Evidence Gaps
08	Project History
09	Traceability & Evidence Coverage
10	Remaining Limitations
11	Technical Appendix
12	Evidence Appendix
13	Bóveda Processing Metadata

The project and Bóveda's position

A concise view of the project, its primary recorded result and the parts of the record that need attention.

What the project does

Chelsea sought to identify housing-related health hazards earlier than reactive enforcement allowed. The project uses linked property and municipal records to predict three categories of code violations and help city government prioritize inspections, although modelling and evaluation cover only previously inspected properties.

Evidence E-B335630551 · E-1352155D41 · E-C113A90389 · E-5CD21A80C3

Best recorded result

PR AUC

0.565

Model

XGBoost

Target

High risk violations

Evaluation

held-out evaluation · 348 observations

Evidence E-C113A90389 · E-B335630551

Main result traced

Bóveda could trace the main result to the data, model and evaluation sample behind it. Some information needed to reproduce the analysis exactly—such as the software setup or data links—is still missing.

What needs attention

6

Signals

3

Evidence Gaps

3 High · 3 Medium

Overall evidence position

The available project record supports some areas more completely than others, but exact reproduction remains incomplete because 3 evidence gaps are still unresolved.

Overall coverage is partial

PROJECT EXPLANATION

Project overview

What the project is intended to do, the objects it analyses and the use it is intended to support.

Purpose

Chelsea sought to identify housing-related health hazards earlier than reactive enforcement allowed. The project uses linked property and municipal records to predict three categories of code violations and help city government prioritize inspections, although modelling and evaluation cover only previously inspected properties.

Evidence E-B335630551 · E-1352155D41 · E-C113A90389 · E-5CD21A80C3

Analytical task	Classify properties by their likelihood of having specified housing-code violations. OBSERVED Evidence E-B335630551
Target / outcome	Binary predictions for any violation, elevated-risk violations, and violations associated with overcrowding. OBSERVED Evidence E-B335630551 · E-06B8ACEBB2
Unit of observation	One Chelsea property or parcel record OBSERVED Evidence E-B335630551 · E-06B8ACEBB2
Population / scope	Chelsea's 5,989 residential properties, with model development restricted to 1,611 inspected properties. OBSERVED Evidence E-B335630551 · E-06B8ACEBB2

Intended use

Support Chelsea city government in prioritizing properties for proactive housing inspections.

OBSERVED

Evidence E-B335630551

Project period

Inspections from 2014–2019; training used proactive inspections from 2014–2018.

OBSERVED

Evidence E-B335630551

Major data sources

BuildingBlocks-linked inspections, police and fire calls, assessor, census, and property records.

OBSERVED

Evidence E-B335630551 · E-06B8ACEBB2

DATA & POPULATIONS

What data the project operated on

Only supervisory questions answered by the available evidence are included below.

Data in brief

Property-level municipal records were joined in BuildingBlocks, filtered to residential properties, transformed into model features, and linked with inspection-derived violation outcomes.

Evidence E-B335630551 · E-06B8ACEBB2

Source data

5,989 residential properties

OBSERVED

Evidence E-B335630551 · E-06B8ACEBB2

Model sample

1,263 training properties

OBSERVED

Evidence E-C113A90389

Evaluation sample

348 held-out properties

OBSERVED

Evidence E-C113A90389

Main population filters

Source records were limited to residential properties; modelling used inspected properties, split by target-area status.

OBSERVED

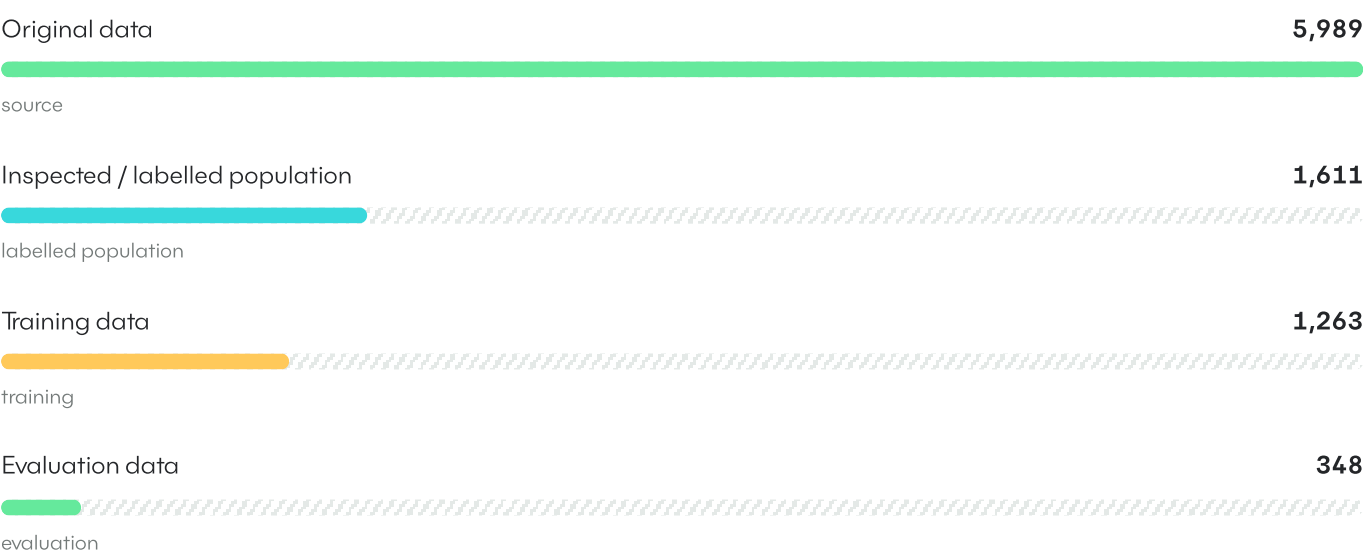
Evidence E-B335630551 · E-06B8ACEBB2 · E-C113A90389

Known population limitation

Evaluation includes only inspected properties outside the target area, so it may not represent Chelsea's uninspected properties.

INTERPRETED_INFERRED

Evidence E-B335630551 · E-C113A90389



Missing data before preparation

Quantified missingness for variables connected to the reconstructed analysis.

VARIABLE	MISSINGNESS	DENOMINATOR	EVIDENCE
Current Use	0%	5,989	OBSERVED Evidence E-06B8ACEBB2
Fuel Type	10.369%	5,989	OBSERVED Evidence E-06B8ACEBB2

VARIABLE	MISSINGNESS	DENOMINATOR	EVIDENCE
Heat Type	10.369%	5,989	OBSERVED Evidence E-06B8ACEBB2
Census Block Group	0.0167%	5,989	OBSERVED Evidence E-06B8ACEBB2
Year Built	8.3012%	6,999	OBSERVED Evidence E-06B8ACEBB2
Building Size	2.7004%	6,999	OBSERVED Evidence E-06B8ACEBB2
Full Bathrooms	9.73%	6,999	OBSERVED Evidence E-06B8ACEBB2
Bedrooms	9.7157%	6,999	OBSERVED Evidence E-06B8ACEBB2
Total Assessed Value	0.7573%	6,999	OBSERVED Evidence E-06B8ACEBB2

Population / sample lineage

How the original population became the training and evaluation samples.

Original data

→ filter →

Inspected / labelled population

Source records were limited to residential properties; modelling used inspected properties, split by target-area status.

Evidence E-B335630551 · E-06B8ACEBB2 · E-C113A90389 · E-1352155D41

Inspected / labelled population

→ split →

Training data

A random forest trained on 1,263 target-area properties was tested on 348 held-out inspected properties outside that area.

Evidence E-C113A90389 · E-1352155D41 · E-5CD21A80C3

Inspected / labelled population

→ split →

Evaluation data

A random forest trained on 1,263 target-area properties was tested on 348 held-out inspected properties outside that area.

Evidence E-C113A90389 · E-1352155D41 · E-5CD21A80C3

What was recorded, and what it supports

The strongest comparable final result is shown with its evaluation context and material limitation.

Best recorded result

PR AUC
0.565

Model	Target	Evaluation sample
XGBoost	High risk violations	held-out evaluation · 348 observations

Evidence E-C113A90389 · E-B335630551

Evaluation design

XGBoost was evaluated for High risk violations on a held-out sample of 348 observations after fitting on 1263 training observations.

DERIVED

Evidence E-C113A90389 · E-B335630551 · E-06B8ACEBB2 · E-1352155D41 · E-5CD21A80C3

Principal limitation

missing from compatible focal evidence.

UNRESOLVED

Other recorded material results

These outcomes retain their own target and evaluation context. They are not treated as directly comparable unless a model comparison below establishes a shared setup.

RESULT	METHOD / TARGET	EVALUATION CONTEXT	EVIDENCE
True positives 59	Default random forest classifier Elevated public-health-risk code violations	Confusion matrix on 348 held-out properties outside the target area	OBSERVED Evidence E-C113A90389
False positives 50	Default random forest classifier Elevated public-health-risk code violations	Confusion matrix on 348 held-out properties outside the target area	OBSERVED Evidence E-C113A90389

What the result establishes

XGBoost achieved 0.565 PR AUC for High risk violations on the held-out evaluation sample of 348 observations.

Evidence E-C113A90389 · E-B335630551 · E-06B8ACEBB2 · E-1352155D41 · E-5CD21A80C3

What it does not establish

missing from compatible focal evidence.

Model comparison

High risk violations

Results recorded for the same target and evaluation setup.

METRIC	XGBOOST	RANDOM FOREST	LASSO REGRESSION	NO SKILL
Average precision calculated from yes/no predictions	0.483202684 3720997	0.463161488 5626494	0.442005402 8285396	—
Accuracy calculated from yes/no predictions	0.65	0.63	0.58	—
Recall calculated from yes/no predictions	0.47	0.4	0.53	—
Precision calculated from yes/no predictions	0.57	0.55	0.48	—
F1 calculated from yes/no predictions	0.52	0.46	0.5	—
ROC AUC	0.678	0.679	0.626	0.5
PR AUC	0.565	0.562	0.504	—

XGBoost records the strongest Average precision (0.4832), while Lasso regression leads on Recall (0.53). No single model leads across every available metric, and the rows should be interpreted separately because they measure different aspects of performance.

Evaluation behaviour

Recorded diagnostics show correct classifications and errors that one headline score can hide.

XGBoost

High risk violations

Rows: actual · columns: predicted

	Predicted 0	Predicted 1
Actual 0	159	49
Actual 1	74	66

For High risk violations, XGBoost correctly identifies 66 of 140 positive cases and misses 74. It also correctly excludes 159 of 208 negative cases while flagging 49 negatives as positive, describing the full error balance in this displayed evaluation sample.

Evidence E-C113A90389

What the displayed model relied on most

The ranking describes relative model reliance within one context; it does not establish cause and effect.



What Bóveda found

Signals describe material project conditions detected by deterministic checks. They do not automatically mean errors, failures, violations or compliance problems.

01

High

High risk violations

The export may use a different prediction field

What Bóveda found

For High risk violations, the code creates one prediction field but the export selects a different field.

Why it matters

The available code does not clearly establish that the exported file contains the prediction intended for that target.

Supported by 1 verification check

Evidence E-06B8ACEBB2 · E-C113A90389

02

High

3 model and target contexts

Evaluation data was used to fit part of the preprocessing

What Bóveda found

For Any violation, preprocessing parameters were fitted using the evaluation data instead of applying a transformation fitted only on the training data. This condition was found in 3 model and target contexts.

Why it matters

The evaluation data influenced part of the model-preparation process, so the reported held-out performance needs additional context.

Supported by 3 verification checks

Evidence E-1352155D41 · E-5CD21A80C3 · E-C113A90389

03

High

Overcrowding · Random Forest

One evaluated model did not identify any positive cases

What Bóveda found

For Overcrowding, Random Forest predicted no positive cases even though 93 positive cases were present in the evaluation sample.

Why it matters

A summary performance score can hide the fact that this model did not identify the positive class in this evaluation.

Supported by 1 verification check

Evidence E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551

04

Medium

Any violation

Different models are used for different parts of the analysis

What Bóveda found

For Any violation, the model with the best recorded result, the model used for feature importance and the model used to produce outputs are not all the same.

Why it matters

The available project record does not clearly document one formally selected model that reconciles these roles.

Supported by 1 verification check

Evidence E-06B8ACEBB2 · E-1352155D41

05

Medium

3 affected target and model contexts

The final model settings differ from the best settings found during tuning

What Bóveda found

For Any violation, the evaluated XGBoost uses different settings from the final recorded best-search configuration. This condition was found in 3 model and target contexts.

Why it matters

The project record does not show that the best tuning configuration was the exact configuration ultimately evaluated.

Supported by 3 verification checks

Evidence E-06B8ACEBB2 · E-1352155D41 · E-5CD21A80C3 · E-B335630551 · E-C113A90389

06

Medium

9 metric calculations

Some Average Precision values were calculated from yes/no predictions

What Bóveda found

For Overcrowding · Random Forest, Average precision was calculated from predicted classes rather than ranked probability scores. This condition was found in 9 model and target contexts.

Why it matters

These values depend on the chosen decision threshold and should not be read as a full measure of ranking performance.

Supported by 9 verification checks

Evidence E-06B8ACEBB2 · E-1352155D41 · E-5CD21A80C3 · E-B335630551 · E-C113A90389

Where the record is incomplete

These are parts of the project that Bóveda could not fully verify or reproduce from the available evidence. A gap does not mean something is wrong; it means the project record is incomplete.

01

Evidence Gap

Reproducibility

Bóveda cannot fully link the predictions to the labels used to evaluate them

What Bóveda found

The available project record does not contain enough information to reconstruct the complete link between predictions and evaluation labels.

Why it matters

Without that link, Bóveda cannot fully reproduce or independently verify the reported evaluation result.

Supported by 1 verification check

Evidence E-06B8ACEBB2 · E-1352155D41 · E-5CD21A80C3 · E-C113A90389

02

Evidence Gap

Reproducibility

The exact software environment cannot be fully reconstructed

What Bóveda found

Some configuration or environment details needed to recreate the analysis are missing from the available project record.

Why it matters

Without these details, Bóveda cannot establish the complete execution setup behind the reported result.

Supported by 1 verification check

Evidence E-06B8ACEBB2 · E-1352155D41 · E-5CD21A80C3 · E-C113A90389

The exact data version used for the result is not fully identified

What Bóveda found

Bóveda found the project data, but not enough information to identify the precise dataset snapshot behind the reported result.

Why it matters

A precise data version is needed to reproduce the same analysis from the same inputs.

Supported by 1 verification check

Evidence E-06B8ACEBB2 · E-1352155D41 · E-5CD21A80C3 · E-C113A90389

PROJECT HISTORY

Material events in the available record

Bóveda found 24 recorded project changes, but only 2 could be connected to a material data, model, evaluation, or release event. The remaining activity is retained as supporting context.

2014–2019



data period

Project data period reconstructed

Inspections from 2014–2019; training used proactive inspections from 2014–2018.

This establishes the period represented by the project evidence; it does not establish when the dataset was created or changed.

Evidence E-B335630551

16 Jun 2020



git commit

added data cleaning notebook

Commit de1f2e2e1b changed 1 file.

No additional rationale was recorded in the commit body.

Evidence E-HIS-BB6B6668B7

13 Jul 2020



git commit

Add files via upload

Commit 063788fb91 changed 3 files.

No additional rationale was recorded in the commit body.

Evidence E-HIS-25C95144A8

Date

unresolved



evaluation

Primary evaluation recorded

recorded project evidence records Average precision score from binary predictions

0.4608713336647534 for Elevated public-health-risk code violations using Default random forest classifier.

The evaluation is supported, but the current project record does not establish when it was performed.

Evidence E-C113A90389

TRACEABILITY & EVIDENCE COVERAGE

How completely Bóveda reconstructed the project

Reconstruction confidence combines the evidence trail behind the main result with whether important applicable analytical areas could be reconstructed. It does not measure model quality, project quality, performance, compliance or governance.

Project reconstruction

Main result traced

Bóveda could trace the main result to the data, model and evaluation sample behind it. Some information needed to reproduce the analysis exactly—such as the software setup or data links—is still missing.

- ✓ The recorded result can be identified.
- ✓ The result is connected to its recorded metric and value.
- ✓ The result is connected to a model and target.
- ✓ The evaluation context can be traced.
- ✓ The evaluation sample and main result can be traced together.
- ✓ Findings retain links to verification checks and Evidence Tokens.

Evidence coverage

Overall coverage is partial

Project Context

Sufficient

Whether the project purpose, scope, methods, decisions and limitations are supported.

Traceability

Not assessed

Whether the main conclusions link to identifiable source material.

Reproducibility

Partial

Whether the main analytical result can be recreated from the available project material.

Source

Sufficient

Whether the source material needed to support the main claims was available for review.

INTERPRETATION BOUNDARY

Remaining limitations

What the client should not conclude from this report, based on the material boundaries represented in the current project record.

- 01 missing from compatible focal evidence.
- 02 Evaluation includes only inspected properties outside the target area, so it may not represent Chelsea's uninspected properties.
- 03 Average precision was calculated from binary class predictions, limiting its interpretation as ranking performance across decision thresholds.
- 04 The evaluation does not establish performance on uninspected Chelsea properties or whether model-

guided inspections improve public-health outcomes.

- 05 The available project record does not contain enough information to reconstruct the complete link between predictions and evaluation labels. Without that link, Bóveda cannot fully reproduce or independently verify the reported evaluation result.
- 06 Some configuration or environment details needed to recreate the analysis are missing from the available project record. Without these details, Bóveda cannot establish the complete execution setup behind the reported result.
- 07 Bóveda found the project data, but not enough information to identify the precise dataset snapshot behind the reported result. A precise data version is needed to reproduce the same analysis from the same inputs.

Absence of evidence is not evidence of absence. These statements are limited to the available project record reviewed by Bóveda.

Technical appendix

Detailed reconstruction retained for technical review. Internal identities and precise analytical terminology are intentionally confined to this layer.

Complete target definitions

TARGET	SOURCE FIELD	CONSTRUCTION	POSITIVE CLASS	INTERPRETATION	STATUS & EVIDENCE
Any violation	any_violation	chelsea.violation_total.mask(chelsea.violation_total > 0, 1)	1	derived binary target	OBSERVED E-1352155D41 · E-06B8ACEBB2
High risk violations	any_high_risk	chelsea.high_risk_violation.mask(chelsea.high_risk_violation > 0, 1)	1	composite target	OBSERVED E-C113A90389 · E-06B8ACEBB2
Overcrowding	any_overcrowding_risk	chelsea.overcrowding_risk.mask(chelsea.overcrowding_risk > 0, 1)	1	proxy target	OBSERVED E-5CD21A80C3 · E-06B8ACEBB2 · E-B335630551

Target-construction ancestry

STEP	TARGET	OPERATION	EXPRESSION / FIELD	STATUS & EVIDENCE
ATCS-D3301A38D1	Target not established		Certificate of Habitability Violations.fillna(0, inplace = True)	OBSERVED E-06B8ACEBB2
ATCS-842BAF0302	Target established		high_risk = ['Certificate of Habitability Violations – Locks', 'Certificate of Habitability Violations – Smoke Detectors and Carbon Monoxide Alarms', 'Certificate of Habitability Violations – Extermination of Insects, Rodents and Skunks', 'Certificate of Habitability Violations – Maintenance of Areas Free from Garbage and Rubbish']	OBSERVED E-06B8ACEBB2
ATCS-CD05F251F3	Target not established		chelsea[high_risk].sum(axis=1)	OBSERVED E-06B8ACEBB2
ATCS-319BA6B267	Target not established		chelsea.high_risk_violation.mask(chelsea.high_risk_violation > 0, 1)	OBSERVED E-06B8ACEBB2
ATCS-B1E9D55FE7	Target not established		chelsea['Certificate of Habitability Violations']	OBSERVED E-06B8ACEBB2
ATCS-CD0025DF51	Target not established		violation_total.fillna(0, inplace = True)	OBSERVED E-06B8ACEBB2
ATCS-602B20F70A	Target not established		chelsea.violation_total.mask(chelsea.violation_total > 0, 1)	OBSERVED E-06B8ACEBB2

STEP	TARGET	OPERATION	EXPRESSION / FIELD	STATUS & EVIDENCE
ATCS-6F538E900C	Target not established		overcrowding_risk = ['Certificate of Habitability Violations - Locks']	OBSERVED E-06B8ACEBB2
ATCS-41588632E5	Target not established		chelsea[overcrowding_risk].sum(axis=1)	OBSERVED E-06B8ACEBB2
ATCS-7AE13CCD4A	Target not established		chelsea.overcrowding_risk.mask(chelsea.overcrowding_risk > 0, 1)	OBSERVED E-06B8ACEBB2
ATCSR C-CE5DCAE153	Target not established	—		OBSERVED E-06B8ACEBB2
ATCSR C-1331083D98	Target not established	—		OBSERVED E-06B8ACEBB2
ATCSR C-7B80D34DA4	Target not established	—		OBSERVED E-06B8ACEBB2
ATCSR C-B2A8C1CE52	Target not established	—		OBSERVED E-06B8ACEBB2
ATCSR C-D1A52E09F7	Target not established	—		OBSERVED E-06B8ACEBB2

STEP	TARGET	OPERATION	EXPRESSION / FIELD	STATUS & EVIDENCE
ATCSR C- 464AC 582C7	Target not establi shed	-		OBSERVE D E- 06B8ACE BB2

Full population and sample table

POPULATION	ROLE	COUNT	UNIT	PREDICATE	STATUS & EVIDENCE
Source rows	source	5,989	observations	–	OBSERVED E-B335630551 · E-06B8ACEBB2
Inspected / labelled population	labelled population	1,611	observations	Source records were limited to residential properties; modelling used inspected properties, split by target-area status.	OBSERVED E-C113A90389 · E-1352155D41
Training population	training	1,263	observations	A random forest trained on 1,263 target-area properties was tested on 348 held-out inspected properties outside that area.	OBSERVED E-C113A90389 · E-1352155D41 · E-5CD21A80C3
Held-out evaluation population	evaluation	348	observations	A random forest trained on 1,263 target-area properties was tested on 348 held-out inspected properties outside that area.	OBSERVED E-C113A90389 · E-1352155D41 · E-5CD21A80C3
Actual scoring population in export code	actual scoring	5,989	observations	Rows supplied to a persisted prediction/export statement	OBSERVED E-1352155D41 · E-C113A90389 · E-5CD21A80C3

Complete model-comparison tables

High risk violations

Results recorded for the same target and evaluation setup.

METRIC	XGBOOST	RANDOM FOREST	LASSO REGRESSION	NO SKILL
Average precision calculated from yes/no predictions	0.483202684 3720997	0.463161488 5626494	0.442005402 8285396	—
Accuracy calculated from yes/no predictions	0.65	0.63	0.58	—
Recall calculated from yes/no predictions	0.47	0.4	0.53	—
Precision calculated from yes/no predictions	0.57	0.55	0.48	—
F1 calculated from yes/no predictions	0.52	0.46	0.5	—
ROC AUC	0.678	0.679	0.626	0.5
PR AUC	0.565	0.562	0.504	—

XGBoost records the strongest Average precision (0.4832), while Lasso regression leads on Recall (0.53). No single model leads across every available metric, and the rows should be interpreted separately because they measure different aspects of performance.

Evidence E-C113A90389

Any violation

Results recorded for the same target and evaluation setup.

METRIC	XGBOOST	RANDOM FOREST	LASSO REGRESSION	NO SKILL
Average precision calculated from yes/no predictions	0.527354111 4058355	0.558165971 9590755	0.504135270 6900464	—
Accuracy calculated from yes/no predictions	0.62	0.66	0.58	—
Recall calculated from yes/no predictions	0.69	0.77	0.76	—
Precision calculated from yes/no predictions	0.56	0.59	0.52	—
F1 calculated from yes/no predictions	0.62	0.67	0.62	—
ROC AUC	0.688	0.706	0.653	0.5
PR AUC	0.608	0.616	0.569	—

Random Forest records the strongest value across all 7 comparable metrics shown for Any violation. The comparison is still row-specific: the values describe different evaluation measures and should not be compared vertically as though they shared one scale.

Evidence E-1352155D41

Overcrowding

Results recorded for the same target and evaluation setup.

METRIC	XGBOOST	RANDOM FOREST	LASSO REGRESSION	NO SKILL
Average precision calculated from yes/no predictions	0.297189184 5640455	0.267241379 3103448	0.298242903 6377868	—
Accuracy calculated from yes/no predictions	0.74	0.73	0.74	—
Recall calculated from yes/no predictions	0.09	0	0.11	—
Precision calculated from yes/no predictions	0.62	0	0.56	—
F1 calculated from yes/no predictions	0.15	0	0.18	—
ROC AUC	0.668	0.635	0.629	0.5
PR AUC	0.427	0.343	0.373	—

XGBoost and Lasso regression share the strongest recorded Accuracy (0.74). No single model leads across every available metric for Overcrowding; each row must be read within its own metric because the scales and meanings differ.

Evidence E-5CD21A80C3

Prevalence and baseline context

TARGET	POPULATION	PREVALENCE	POSITIVE / TOTAL	EVIDENCE
High risk violations	Training population	0.470309	594 / 1263	E-C113A90389
High risk violations	Held-out evaluation population	0.402299	140 / 348	E-C113A90389
Overcrowding	Training population	0.298496	377 / 1263	E-5CD21A80C3
Overcrowding	Held-out evaluation population	0.267241	93 / 348	E-5CD21A80C3
Any violation	Inspected / labelled population	0.534451	861 / 1611	E-06B8ACEBB2
High risk violations	Inspected / labelled population	0.455618	734 / 1611	E-06B8ACEBB2

TARGET	POPULATION	PREVALENCE	POSITIVE / TOTAL	EVIDENCE
Overcrowding	Inspected / labelled population	0.291744	470 / 1611	E-06B8ACEBB2
Any violation	Held-out evaluation population	0.448276	156 / 348	E-1352155D41
Any violation	Training population	0.558195	705 / 1263	E-06B8ACEBB2 · E-1352155D41

Additional evaluation diagnostics

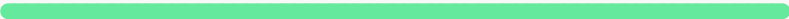
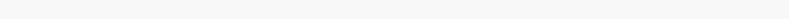
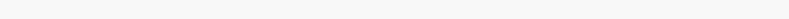
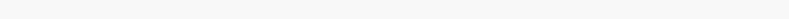
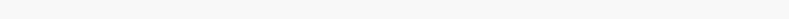
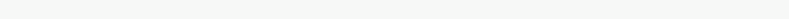
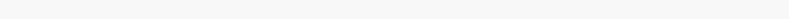
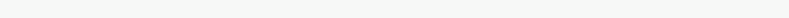
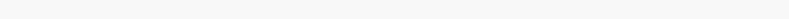
TARGET	METHOD	DIAGNOSTIC	PHASE	PERSISTED VALUE	EVIDENCE
High risk violations	Lasso regression	confusion matrix	test	[[127,81],[66,74]]	E-C113A90389
High risk violations	Random Forest	confusion matrix	test	[[163,45],[84,56]]	E-C113A90389
High risk violations	XGBoost	confusion matrix	test	[[159,49],[74,66]]	E-C113A90389
High risk violations	XGBoost	confusion matrix	test	[[155,53],[81,59]]	E-C113A90389
High risk violations	Random Forest	confusion matrix	test	[[158,50],[81,59]]	E-C113A90389
Any violation	Lasso regression	confusion matrix	test	[[84,108],[38,118]]	E-1352155D41
Overcrowding	Lasso regression	confusion matrix	test	[[247,8],[83,10]]	E-5CD21A80C3
Any violation	Random Forest	confusion matrix	test	[[109,83],[36,120]]	E-1352155D41
Overcrowding	Random Forest	confusion matrix	test	[[255,0],[93,0]]	E-5CD21A80C3
Any violation	XGBoost	confusion matrix	test	[[108,84],[48,108]]	E-1352155D41
Overcrowding	XGBoost	confusion matrix	test	[[250,5],[85,8]]	E-5CD21A80C3
Any violation	Random Forest	predicted positive count	test	162	E-1352155D41
Any violation	Random Forest	predicted positive count	test	203	E-1352155D41
Any violation	XGBoost	predicted positive count	test	192	E-1352155D41
Any violation	Lasso regression	predicted positive count	test	348	E-1352155D41
Any violation	Lasso regression	predicted positive count	test	226	E-1352155D41
High risk violations	Random Forest	predicted positive count	test	109	E-C113A90389
High risk violations	Random Forest	predicted positive count	test	101	E-C113A90389
High risk violations	Random Forest	predicted positive count	test	101	E-C113A90389
High risk violations	XGBoost	predicted positive count	test	115	E-C113A90389
High risk violations	Lasso regression	predicted positive count	test	0	E-C113A90389

TARGET	METHOD	DIAGNOSTIC	PHASE	PERSISTED VALUE	EVIDENCE
High risk violations	Lasso regression	predicted positive count	test	155	E-C113A90389
Overcrowding	Random Forest	predicted positive count	test	39	E-5CD21A80C3
Overcrowding	Random Forest	predicted positive count	test	0	E-5CD21A80C3
Overcrowding	XGBoost	predicted positive count	test	32	E-5CD21A80C3
Overcrowding	XGBoost	predicted positive count	test	13	E-5CD21A80C3
Overcrowding	Lasso regression	predicted positive count	test	0	E-5CD21A80C3
Overcrowding	Lasso regression	predicted positive count	test	18	E-5CD21A80C3

Full feature-evidence sets

High risk violations

XGBoost · final

Tract 160101 Block Group 5		0.051735
Condo NL MDL-05		0.046304
Tract 160501 Block Group 2		0.032077
APT 4-UNT MDL-94		0.028946
Heat: Steam		0.025094
Tract 160502 Block Group 3		0.024519
Current Use category · unresolved label		0.02424
Opportunityzone: None		0.023925
Tract 160101 Block Group 4		0.023181
Tract 160101 Block Group 2		0.021499
Opportunityzone: Tract 160200		0.020858
Tract 160101 Block Group 1		0.019874
Vacant Foreclosed Properties		0.019032
Tract 160101 Block Group 3		0.018658
Tract 160200 Block Group 1		0.018541
Tract 160200 Block Group 2		0.018508
Tract 160501 Block Group 3		0.018213

logbuildingsize		0.017545
Tract 160501 Block Group 5		0.017382
Rat Baiting - Count		0.017193

For High risk violations, the displayed XGBoost model relies most on Tract 160101 Block Group 5 (0.0517) and Condo NL MDL-05 (0.0463). These are the two highest recorded importance values in this displayed context; the ranking describes model reliance, not whether either input drives the outcome.

Feature importance shows model reliance, not cause and effect.

Evidence E-C113A90389

Any violation

Random Forest · final

logbuildingsize		0.061521
Building Size		0.059653
Total Assessed Value		0.058384
Land Size		0.057681
logassessedvalue		0.054454
40U Violations - Count		0.044459
Property Sales - Price		0.042202
40U Violations - Fine		0.041145
Fire Incidents - Count		0.034114
Year Built		0.034032
Bedrooms		0.032387
Police Incidents - Count		0.031724
logbedrooms		0.029288
Police Incidents After 2015		0.027598
Tract 160101 Block Group 5		0.024083
Property Sales - Count		0.023782
Building Permits - Count		0.021821
Rat Baiting - Count		0.018769
logbathrooms		0.018151
Full Bathrooms		0.017003

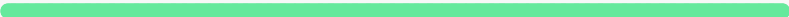
For Any violation, the displayed Random Forest model relies most on logbuildingsize (0.0615) and Building Size (0.0597). These are the two highest recorded importance values in this displayed context; the ranking describes model reliance, not whether either input drives the outcome.

Feature importance shows model reliance, not cause and effect.

Evidence E-1352155D41

Overcrowding

XGBoost · final

Condo NL MDL-05		0.047446
Opportunityzone: None		0.03134
Opportunityzone: Tract 160200		0.030647
Tract 160101 Block Group 5		0.030643
Three-Family Residential		0.029607
Tract 160501 Block Group 2		0.026508
Tract 160101 Block Group 1		0.023954
Tract 160501 Block Group 4		0.02303
Foreclosure Sales - Count		0.02258
Tract 160101 Block Group 4		0.021577
Tax Title		0.020767
Tract 160101 Block Group 2		0.020703
logbuildingsize		0.019527
APT OVER 8 MDL-94		0.019289
Tract 160101 Block Group 3		0.019273
Land Size		0.019189
Building Size		0.019169
Heat: Forced Air-Duc		0.019004
Apartments with Four to Eight Units		0.01889
Tract 160200 Block Group 2		0.018509

For Overcrowding, the displayed XGBoost model relies most on Condo NL MDL-05 (0.0474) and Opportunityzone: None (0.0313). These are the two highest recorded importance values in this displayed context; the ranking describes model reliance, not whether either input drives the outcome.

Feature importance shows model reliance, not cause and effect.

Evidence E-5CD21A80C3

Tuning and search stages

Target	Method	Search	CV	Candidates	Best Score	Final Relation	Evidence
Any violation	Random Forest	GridSearchCV · stage 1	5	1152	0.742484	parameter mismatch	E-1352155D41
Any violation	Random Forest	GridSearchCV · stage 2	5	80	0.742484	parameter mismatch	E-1352155D41
Any violation	Random Forest	GridSearchCV · stage 3	5	30	0.742619	parameters match shared best	E-1352155D41
Any violation	XGBoost	GridSearchCV · stage 1	5	21	0.728367	parameter mismatch	E-1352155D41
Any violation	XGBoost	GridSearchCV · stage 2	5	15	0.729934	parameter mismatch	E-1352155D41
Any violation	XGBoost	GridSearchCV · stage 3	5	2400	0.737626	parameter mismatch	E-1352155D41
Any violation	XGBoost	GridSearchCV · stage 4	5	36	0.739808	parameter mismatch	E-1352155D41
Any violation	XGBoost	GridSearchCV · stage 5	5	25	0.739808	parameter mismatch	E-1352155D41
Any violation	Lasso regression	manual cross validation · stage 1	5	30	62.861585	linked by search variable	E-1352155D41
Any violation	Lasso regression	manual cross validation · stage 2	5	30	64.707612	linked by search variable	E-1352155D41
High risk violations	Random Forest	GridSearchCV · stage 1	5	1152	0.654105	parameter mismatch	E-C113A90389
High risk violations	Random Forest	GridSearchCV · stage 2	5	36	0.654105	parameter mismatch	E-C113A90389
High risk violations	Random Forest	GridSearchCV · stage 3	5	9	0.654278	parameters match shared best	E-C113A90389
High risk violations	XGBoost	GridSearchCV · stage 1	5	21	0.623302	parameter mismatch	E-C113A90389
High risk violations	XGBoost	GridSearchCV · stage 2	5	12	0.624294	parameter mismatch	E-C113A90389
High risk violations	XGBoost	GridSearchCV · stage 3	5	1344	0.639879	parameter mismatch	E-C113A90389
High risk violations	XGBoost	GridSearchCV · stage 4	5	48	0.610458	parameter mismatch	E-C113A90389
High risk violations	XGBoost	GridSearchCV · stage 5	5	16	0.610458	parameter mismatch	E-C113A90389
High risk violations	Lasso regression	manual cross validation · stage 1	5	30	55.251671	linked by search variable	E-C113A90389
High risk violations	Lasso regression	manual cross validation · stage 2	5	30	55.526334	linked by search variable	E-C113A90389
Overcrowding	Random Forest	GridSearchCV · stage 1	5	1152	0.438829	parameter mismatch	E-5CD21A80C3

TARGET	METHOD	SEARCH	CV	CANDIDATES	BEST SCORE	FINAL RELATION	EVIDENCE
Overcrowding	Random Forest	GridSearchCV · stage 2	5	27	0.43514	parameter mismatch	E-5CD21A80C3
Overcrowding	Random Forest	GridSearchCV · stage 3	5	18	0.43514	parameters match shared best	E-5CD21A80C3
Overcrowding	XGBoost	GridSearchCV · stage 1	5	21	0.411544	parameter mismatch	E-5CD21A80C3
Overcrowding	XGBoost	GridSearchCV · stage 2	5	18	0.411544	parameter mismatch	E-5CD21A80C3
Overcrowding	XGBoost	GridSearchCV · stage 3	5	2400	0.418027	parameter mismatch	E-5CD21A80C3
Overcrowding	XGBoost	GridSearchCV · stage 4	5	16	0.418027	parameter mismatch	E-5CD21A80C3
Overcrowding	Lasso regression	manual cross validation · stage 1	5	30	34.319513	linked by search variable	E-5CD21A80C3
Overcrowding	Lasso regression	manual cross validation · stage 2	5	30	34.374653	linked by search variable	E-5CD21A80C3

Model variants

TARGET	METHOD	ESTIMATOR	ROLES	PARAMETERS	SOURCE	EVIDENCE
High risk violations	Random Forest	RandomForestClassifier	candidate final	random_state = 1	models/Predicting high risk violations.ipynb#cell-21	E-C113A90389
High risk violations	Random Forest	RandomForestClassifier	final evaluated	max_depth = 15, n_estimators=550, max_features='sqrt', min_samples_leaf=1, min_samples_split=2, criterion='gini', random_state=1	models/Predicting high risk violations.ipynb#cell-27	E-C113A90389
High risk violations	XGBoost	XGBClassifier	untuned, held out untuned	random_state=1	models/Predicting high risk violations.ipynb#cell-29	E-C113A90389
High risk violations	XGBoost	XGBClassifier	final evaluated, feature producing, export producing	n_estimators = 75, max_depth = 12, min_child_weight = 1, colsample_bytree = 0.6, subsample = 0.5, gamma = 0.3, alpha = 0.3, reg_lambda = 0.3, learning_rate = 0.01, random_state=1	models/Predicting high risk violations.ipynb#cell-41	E-C113A90389
High risk violations	Lasso regression	Lasso	untuned, whole population scoring	{}	models/Predicting high risk violations.ipynb#cell-43	E-C113A90389
High risk violations	Lasso regression	Lasso	candidate final, whole population scoring	alpha = i	models/Predicting high risk violations.ipynb#cell-44	E-C113A90389
High risk violations	Lasso regression	Lasso	candidate final, whole population scoring	alpha = i	models/Predicting high risk violations.ipynb#cell-45	E-C113A90389
High risk violations	Lasso regression	Lasso	final evaluated, whole population scoring	alpha = best_i	models/Predicting high risk violations.ipynb#cell-46	E-C113A90389
Overcrowding	Random Forest	RandomForestClassifier	untuned, held out untuned	random_state=1	models/Predicting overcrowding.ipynb#cell-21	E-5CD21A80C3
Overcrowding	Random Forest	RandomForestClassifier	candidate final	random_state = 1	models/Predicting overcrowding.ipynb#cell-22	E-5CD21A80C3

TARGET	METHOD	ESTIMATOR	ROLES	PARAMETERS	SOURCE	EVIDENCE
Overcrowding	Random Forest	RandomForestClassifier	final evaluated	max_depth = 18, n_estimators=25, max_features='log2', min_samples_leaf=2, min_samples_split=100, criterion='gini', random_state=1	models/Predicting overcrowding.ipynb#cell-28	E-5CD21A80C3
Overcrowding	XGBoost	XGBClassifier	untuned, held out untuned	random_state=1	models/Predicting overcrowding.ipynb#cell-30	E-5CD21A80C3
Overcrowding	XGBoost	XGBClassifier	final evaluated, feature producing, export producing	n_estimators = 250, max_depth = 5, min_child_weight = 1, colsample_bytree = 0.5, subsample = 0.75, gamma = 0.0, alpha = 0.3, reg_lambda = 0.3, learning_rate = 0.01, random_state=1	models/Predicting overcrowding.ipynb#cell-39	E-5CD21A80C3
Overcrowding	Lasso regression	Lasso	untuned, whole population scoring	{}	models/Predicting overcrowding.ipynb#cell-41	E-5CD21A80C3
Overcrowding	Lasso regression	Lasso	candidate final, whole population scoring	alpha = i	models/Predicting overcrowding.ipynb#cell-42	E-5CD21A80C3
Overcrowding	Lasso regression	Lasso	candidate final, whole population scoring	alpha = i	models/Predicting overcrowding.ipynb#cell-43	E-5CD21A80C3
Overcrowding	Lasso regression	Lasso	final evaluated, whole population scoring	alpha = best_i	models/Predicting overcrowding.ipynb#cell-44	E-5CD21A80C3

Selection statements

STATEMENT	TARGET	METHOD	METRIC	VALUE	STATUS	EVIDENCE
best final on metric	High risk violations	XGBoost	average_precision	0.483203	established	E-C113A90389
best final on metric	High risk violations	XGBoost	accuracy	0.65	established	E-C113A90389
best final on metric	High risk violations	Lasso regression	recall	0.53	established	E-C113A90389
best final on metric	High risk violations	XGBoost	precision	0.57	established	E-C113A90389
best final on metric	High risk violations	XGBoost	f1	0.52	established	E-C113A90389
best final on metric	High risk violations	Random Forest	roc_auc	0.679	established	E-C113A90389
best final on metric	High risk violations	XGBoost	pr_auc	0.565	established	E-C113A90389
best final on metric	Any violation	Random Forest	average_precision	0.558166	established	E-1352155D41
best final on metric	Any violation	Random Forest	accuracy	0.66	established	E-1352155D41
best final on metric	Any violation	Random Forest	recall	0.77	established	E-1352155D41
best final on metric	Any violation	Random Forest	precision	0.59	established	E-1352155D41
best final on metric	Any violation	Random Forest	f1	0.67	established	E-1352155D41
best final on metric	Any violation	Random Forest	roc_auc	0.706	established	E-1352155D41
best final on metric	Any violation	Random Forest	pr_auc	0.616	established	E-1352155D41
best final on metric	Overcrowding	Lasso regression	average_precision	0.298243	established	E-5CD21A80C3
best final on metric	Overcrowding	XGBoost	accuracy	0.74	established	E-5CD21A80C3
best final on metric	Overcrowding	Lasso regression	recall	0.11	established	E-5CD21A80C3
best final on metric	Overcrowding	XGBoost	precision	0.62	established	E-5CD21A80C3
best final on metric	Overcrowding	Lasso regression	f1	0.18	established	E-5CD21A80C3
best final on metric	Overcrowding	XGBoost	roc_auc	0.668	established	E-5CD21A80C3
best final on metric	Overcrowding	XGBoost	pr_auc	0.427	established	E-5CD21A80C3

STATEMENT	TARGET	METHOD	METRIC	VALUE	STATUS	EVIDENCE
explicit best label	Any violation	–	–	–	established	E-1352155D41
export producing model	Any violation	XGBoost	–	–	established	E-1352155D41
export producing model	High risk violations	XGBoost	–	–	established	E-C113A90389
export producing model	Overcrowding	XGBoost	–	–	established	E-5CD21A80C3
project selected model	Target not established	–	–	–	unresolved	

Output relationships

TARGET	PRODUCER	OUTPUT	PATH	EXISTENCE	CONSISTENCY	EVIDENCE
Target not established	Unresolved	analytical data export	chelsea_all.csv	referenced not persisted	no structural mismatch detected	E-06B8ACEBB2
Target not established	Unresolved	analytical data export	chelsea_clean.csv	referenced not persisted	no structural mismatch detected	E-06B8ACEBB2
Any violation	XGBoost	prediction export	final_predictions_any_violatins.csv	referenced not persisted	no structural mismatch detected	E-1352155D41
High risk violations	XGBoost	prediction export	final_predictions_high_risk.csv	referenced not persisted	assigned column not selected	E-C113A90389
Overcrowding	XGBoost	prediction export	final_predictions_overcrowding.csv	referenced not persisted	no structural mismatch detected	E-5CD21A80C3

Scoped limitations

KIND	SUBJECT	OBSERVED FACT	DERIVED RELATION	INTERPRETATION BOUNDARY	EVIDENCE
preprocessing fit scope	DataFormati onStage	scaler is fitted separately on training and evaluation inputs.	One fitted preprocessing state is not shared across the split.	Evaluation inputs are transformed with parameters learned from the evaluation population.	E-1352155D41 · E- C113A90389 · E- 5CD21A80C3
hard label average precision	MetricObser vation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-1352155D41
hard label average precision	MetricObser vation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-1352155D41
hard label average precision	MetricObser vation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-1352155D41
hard label average precision	MetricObser vation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-C113A90389
hard label average precision	MetricObser vation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-C113A90389
hard label average precision	MetricObser vation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-C113A90389
hard label average precision	MetricObser vation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-5CD21A80C3
hard label average precision	MetricObser vation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-5CD21A80C3
hard label average precision	MetricObser vation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-5CD21A80C3
hard label average precision	MetricObser vation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-1352155D41

KIND	SUBJECT	OBSERVED FACT	DERIVED RELATION	INTERPRETATION BOUNDARY	EVIDENCE
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-1352155D41
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-1352155D41
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-1352155D41
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-1352155D41
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-1352155D41
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-C113A90389
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-C113A90389
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-C113A90389
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-C113A90389
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-C113A90389

KIND	SUBJECT	OBSERVED FACT	DERIVED RELATION	INTERPRETATION BOUNDARY	EVIDENCE
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-5CD21A80C3
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-5CD21A80C3
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-5CD21A80C3
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-5CD21A80C3
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-5CD21A80C3
hard label average precision	MetricObservation	average_precision_score receives hard class predictions.	The observation is threshold-dependent rather than a ranking summary.	It does not by itself establish prioritisation quality across ranked properties.	E-5CD21A80C3
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-1352155D41
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-1352155D41
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-1352155D41
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-1352155D41
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-1352155D41

KIND	SUBJECT	OBSERVED FACT	DERIVED RELATION	INTERPRETATION BOUNDARY	EVIDENCE
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-1352155D41
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-1352155D41
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-C113A90389
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-C113A90389
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-C113A90389
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-C113A90389
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-C113A90389
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-C113A90389
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-C113A90389
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-5CD21A80C3
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-5CD21A80C3

KIND	SUBJECT	OBSERVED FACT	DERIVED RELATION	INTERPRETATION BOUNDARY	EVIDENCE
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-5CD21A80C3
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-5CD21A80C3
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-5CD21A80C3
search final parameter mismatch	HyperparameterSearch	Persisted best parameters and the related final estimator parameters differ.	The evaluated final estimator is not an exact instantiation of this search stage's best parameters.	A model-selection rationale cannot be inferred from notebook order.	E-5CD21A80C3
output column inconsistency	OutputProductionStatement	Assigned prediction column and selected export columns are inconsistent (assigned_column_not_selected).	The persisted export statement does not select the assigned prediction column.	Successful creation of the intended target output is not established.	E-C113A90389
proxy target	TargetDefinition	Persisted project evidence relates Overcrowding to a broader construct represented by its recorded source indicator.	–	The prediction target should be presented as an indicator or proxy rather than a direct measurement of the broader construct.	E-5CD21A80C3 · E-06B8ACEBB2 · E-B335630551

Checks performed

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-CURRENT-PURPOSE-TRACE	Current purpose source	Not detected	implemented	Current purpose source is resolved to the inspected evidence.	E-B335630551
CHK-PROJECT-CONTEXT-COVERAGE	Material project context	Not detected	implemented	Material project context is resolved to the inspected evidence.	E-B335630551 · E-06B8ACEBB2
CHK-SOURCE-INVENTORY-COVERAGE	Authorised source inventory	Not detected	implemented	Authorised source inventory is resolved to the inspected evidence.	E-B335630551 · E-06B8ACEBB2
CHK-TYPED-FOCAL-RESULT-ORIGIN	Typed focal-result origin	Not detected	implemented graph native	The focal result resolves through compatible evidence-bound target, run, evaluation, metric, and population objects.	E-06B8ACEBB2 · E-1352155D41 · E-5CD21A80C3 · E-C113A90389
CHK-TARGET-CONSTRUCTION-ANCESTRY	Complete target construction ancestry	Not detected	implemented graph native	Any violation terminates through a complete evidence-bound construction ancestry.	E-06B8ACEBB2 · E-1352155D41
CHK-TARGET-CONSTRUCTION-ANCESTRY	Complete target construction ancestry	Not detected	implemented graph native	High risk violations terminates through a complete evidence-bound construction ancestry.	E-06B8ACEBB2 · E-C113A90389
CHK-TARGET-CONSTRUCTION-ANCESTRY	Complete target construction ancestry	Not detected	implemented graph native	Overcrowding terminates through a complete evidence-bound construction ancestry.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-POPULATION-LINEAGE-COMPATIBILITY	Population lineage compatibility	Not enough evidence	implemented graph native	The population lineage is missing an original, training or evaluation population, a connecting relationship, or its supporting evidence.	E-06B8ACEBB2 · E-1352155D41 · E-5CD21A80C3 · E-B335630551 · E-C113A90389
CHK-EVALUATION-INTENDED-POPULATION-ALIGNMENT	Evaluation versus intended prediction population	Not applicable	implemented graph native	No population-specific intended prediction or decision population is represented.	
CHK-EVALUATION-PREPROCESSOR-FIT-SCOPE	Evaluation preprocessing fit scope	Condition found	implemented graph native	A fitted preprocessing transformation for Any violation estimates parameters from held-out evaluation inputs.	E-1352155D41
CHK-EVALUATION-PREPROCESSOR-FIT-SCOPE	Evaluation preprocessing fit scope	Condition found	implemented graph native	A fitted preprocessing transformation for High risk violations estimates parameters from held-out evaluation inputs.	E-C113A90389
CHK-EVALUATION-PREPROCESSOR-FIT-SCOPE	Evaluation preprocessing fit scope	Condition found	implemented graph native	A fitted preprocessing transformation for Overcrowding estimates parameters from held-out evaluation inputs.	E-5CD21A80C3

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible continuous score input.	E-06B8ACEBB2 · E-1352155D41
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible probability input.	E-06B8ACEBB2 · E-1352155D41
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible probability input.	E-06B8ACEBB2 · E-1352155D41
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible constant score input.	E-06B8ACEBB2 · E-1352155D41
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	PR AUC uses a compatible probability input.	E-06B8ACEBB2 · E-1352155D41
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	PR AUC uses a compatible probability input.	E-06B8ACEBB2 · E-1352155D41
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	PR AUC uses a compatible probability input.	E-06B8ACEBB2 · E-1352155D41
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible continuous score input.	E-06B8ACEBB2 · E-C113A90389
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible probability input.	E-06B8ACEBB2 · E-C113A90389
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible probability input.	E-06B8ACEBB2 · E-C113A90389

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible constant score input.	E-06B8ACEBB2 · E-C113A90389
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	PR AUC uses a compatible probability input.	E-06B8ACEBB2 · E-C113A90389
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	PR AUC uses a compatible probability input.	E-06B8ACEBB2 · E-C113A90389
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	PR AUC uses a compatible probability input.	E-06B8ACEBB2 · E-C113A90389
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible continuous score input.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible probability input.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible probability input.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	ROC AUC uses a compatible constant score input.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	PR AUC uses a compatible probability input.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	PR AUC uses a compatible probability input.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not detected	implemented graph native	PR AUC uses a compatible probability input.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Condition found	implemented graph native	Average precision for Any violation · Random Forest is computed from hard class predictions and is threshold-dependent.	E-06B8ACEBB2 · E-1352155D41
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Condition found	implemented graph native	Average precision for Any violation · XGBoost is computed from hard class predictions and is threshold-dependent.	E-06B8ACEBB2 · E-1352155D41
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Condition found	implemented graph native	Average precision for Any violation · Lasso regression is computed from hard class predictions and is threshold-dependent.	E-06B8ACEBB2 · E-1352155D41
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Condition found	implemented graph native	Average precision for High risk violations · Random Forest is computed from hard class predictions and is threshold-dependent.	E-06B8ACEBB2 · E-C113A90389
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Condition found	implemented graph native	Average precision for High risk violations · XGBoost is computed from hard class predictions and is threshold-dependent.	E-06B8ACEBB2 · E-C113A90389
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Condition found	implemented graph native	Average precision for High risk violations · Lasso regression is computed from hard class predictions and is threshold-dependent.	E-06B8ACEBB2 · E-C113A90389
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Condition found	implemented graph native	Average precision for Overcrowding · Random Forest is computed from hard class predictions and is threshold-dependent.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Condition found	implemented graph native	Average precision for Overcrowding · XGBoost is computed from hard class predictions and is threshold-dependent.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Condition found	implemented graph native	Average precision for Overcrowding · Lasso regression is computed from hard class predictions and is threshold-dependent.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Not detected	implemented graph native	Lasso regression produced positive predictions in this held-out diagnostic.	E-06B8ACEBB2 · E-C113A90389

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Not detected	implemented graph native	Random Forest produced positive predictions in this held-out diagnostic.	E-06B8ACEBB2 · E-C113A90389
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Not detected	implemented graph native	XGBoost produced positive predictions in this held-out diagnostic.	E-06B8ACEBB2 · E-C113A90389
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Not detected	implemented graph native	Lasso regression produced positive predictions in this held-out diagnostic.	E-06B8ACEBB2 · E-1352155D41
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Not detected	implemented graph native	Lasso regression produced positive predictions in this held-out diagnostic.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Not detected	implemented graph native	Random Forest produced positive predictions in this held-out diagnostic.	E-06B8ACEBB2 · E-1352155D41
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Condition found	implemented graph native	Random Forest for Overcrowding produced no positive predictions despite 93 actual positive cases.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Not detected	implemented graph native	XGBoost produced positive predictions in this held-out diagnostic.	E-06B8ACEBB2 · E-1352155D41
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Not detected	implemented graph native	XGBoost produced positive predictions in this held-out diagnostic.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Not detected	implemented graph native	The evaluated final estimator corresponds to the terminal recorded best-search parameters.	E-06B8ACEBB2 · E-1352155D41
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Condition found	implemented graph native	XGBoost for Any violation uses explicit parameters different from the terminal recorded best-search result.	E-06B8ACEBB2 · E-1352155D41
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Not detected	implemented graph native	The final estimator is reconstructibly linked to the terminal selected search value.	E-06B8ACEBB2 · E-1352155D41
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Not detected	implemented graph native	The evaluated final estimator corresponds to the terminal recorded best-search parameters.	E-06B8ACEBB2 · E-C113A90389
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Condition found	implemented graph native	XGBoost for High risk violations uses explicit parameters different from the terminal recorded best-search result.	E-06B8ACEBB2 · E-C113A90389
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Not detected	implemented graph native	The final estimator is reconstructibly linked to the terminal selected search value.	E-06B8ACEBB2 · E-C113A90389

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Not detected	implemented graph native	The evaluated final estimator corresponds to the terminal recorded best-search parameters.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Condition found	implemented graph native	XGBoost for Overcrowding uses explicit parameters different from the terminal recorded best-search result.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Not detected	implemented graph native	The final estimator is reconstructibly linked to the terminal selected search value.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-OUTPUT-COLUMN-CONSISTENCY	Output column consistency	Not detected	implemented graph native	The prediction output for Any violation selects its assigned prediction column.	E-06B8ACEBB2 · E-1352155D41
CHK-OUTPUT-COLUMN-CONSISTENCY	Output column consistency	Condition found	implemented graph native	The prediction output for High risk violations does not select the column assigned for that same target.	E-06B8ACEBB2 · E-C113A90389
CHK-OUTPUT-COLUMN-CONSISTENCY	Output column consistency	Not detected	implemented graph native	The prediction output for Overcrowding selects its assigned prediction column.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-ANALYTICAL-ROLE-ALIGNMENT	Analytical model-role alignment	Condition found	implemented graph native	Best-result, feature-producing, and output-producing roles for Any violation refer to different model families while formal selection remains unresolved.	E-06B8ACEBB2 · E-1352155D41
CHK-ANALYTICAL-ROLE-ALIGNMENT	Analytical model-role alignment	Not detected	implemented graph native	Material analytical roles for High risk violations align or are resolved by an explicit project-selection statement.	E-06B8ACEBB2 · E-C113A90389
CHK-ANALYTICAL-ROLE-ALIGNMENT	Analytical model-role alignment	Not detected	implemented graph native	Material analytical roles for Overcrowding align or are resolved by an explicit project-selection statement.	E-06B8ACEBB2 · E-5CD21A80C3 · E-B335630551
CHK-REPRODUCTION-INPUT-COVERAGE-V2	Result reproduction inputs	Not enough evidence	implemented graph native	One or more independent reproduction-input categories are unresolved in the available project record.	E-06B8ACEBB2 · E-1352155D41 · E-5CD21A80C3 · E-C113A90389
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Main result	Not detected	implemented analytical area native	Main result answers its defined supervisory question with connected evidence.	E-C113A90389 · E-06B8ACEBB2 · E-B335630551 · E-1352155D41 · E-5CD21A80C3
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Model comparison	Not detected	implemented analytical area native	Model comparison answers its defined supervisory question with connected evidence.	E-C113A90389 · E-06B8ACEBB2 · E-B335630551

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Feature / driver evidence	Not detected	implemented analytical area native	Feature / driver evidence answers its defined supervisory question with connected evidence.	E-C113A90389 · E-06B8ACEBB2 · E-B335630551
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Evaluation behaviour	Not detected	implemented analytical area native	Evaluation behaviour answers its defined supervisory question with connected evidence.	E-C113A90389 · E-06B8ACEBB2 · E-B335630551
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Population and sample lineage	Not detected	implemented analytical area native	Population and sample lineage answers its defined supervisory question with connected evidence.	E-C113A90389 · E-06B8ACEBB2 · E-B335630551 · E-1352155D41 · E-5CD21A80C3
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Missing data before preparation	Not applicable	implemented analytical area native	Missing data before preparation is not demonstrably applicable to the reconstructed main-target story.	E-C113A90389 · E-06B8ACEBB2 · E-B335630551
CHK-MATERIAL-EVIDENCE-ABSENCE	Material evidence absence	Not detected	implemented finding native	No applicable analytical area is entirely unavailable because of material evidence absence.	

Complete History projection

DATE	TYPE	EVENT	DESCRIPTION	ROLE	EVIDENCE
2014–2019	data period	Project data period reconstructed	Inspections from 2014–2019; training used proactive inspections from 2014–2018.	Material	E-B335630551
16 Jun 2020	git commit	added data cleaning notebook	Commit de1f2e2e1b changed 1 file.	Material	E-HIS-BB6B6668B7
13 Jul 2020	git commit	Add files via upload	Commit 063788fb91 changed 3 files.	Material	E-HIS-25C95144A8
Date unresolved	evaluation	Primary evaluation recorded	Persisted project evidence records Average precision score from binary predictions 0.4608713336647534 for Elevated public-health-risk code violations using Default random forest classifier.	Material	E-C113A90389
26 Aug 2026, 11:52 UTC	boveda audit	Bóveda audit completed	Bóveda completed audit AUD-A57071CBE3 and stored the current canonical Project Record. 2 reconstruction attempts were recorded; 1 did not pass validation and 1 triggered a retry.	Material	

The PDF retains material History rows. 22 supporting project changes remain available in the complete HTML report.

Technical provenance

ANALYTICAL GRAPH SCHEMA boveda-analytical-object-graph-0.13.2	DERIVATION "deterministic_persisted_evidence_v0.13.2"
COMPLETENESS <pre>{"targets":"available","populations":"available","model_runs":"available","searches":"available","metrics":"available","diagnostics":"available","prevalence":"available","outputs":"available","evidence_termination":"complete"}</pre>	PROVIDER CALLS USED BY THIS ANALYTICAL PROJECTION 0

Evidence appendix

A stable index linking material report claims to source evidence retained in the Project Record.

Evidence Index

EVIDENCE TOKEN	SOURCE	SOURCE LOCATOR	EPISTEMIC STATUS	SHORT DESCRIPTION
E-06B8ACEBB2	data-cleaning/data-cleaning.ipynb	data-cleaning/data-cleaning.ipynb	OBSERVED	[CODE FOR OUTPUT] print(chelsea.shape) print(chelsea.columns) [PERSISTED OUTPUT] (7036, 68) Index(['Parcel ID', 'Address', 'Building Permits - Count - Between 01-01-2016 and 05-31-2019',...
E-1352155D41	models/Predicting any code violation.ipynb	models/Predicting any code violation.ipynb	OBSERVED	[CODE FOR OUTPUT] chelsea = chelsea_df.copy() chelsea.shape [PERSISTED OUTPUT] (1611, 109) [CODE FOR OUTPUT] chelsea.head(2) [PERSISTED OUTPUT] Building Permits - Count See Click Fix - Co...
E-5CD21A80C3	models/Predicting overcrowding.ipynb	models/Predicting overcrowding.ipynb	OBSERVED	[CODE FOR OUTPUT] chelsea = chelsea_df.copy() chelsea.shape [PERSISTED OUTPUT] (1611, 109) [CODE FOR OUTPUT] chelsea.head(2) [PERSISTED OUTPUT] Building Permits - Count See Click Fix - Co...
E-B335630551	README.md	README.md	OBSERVED	# Leveraging data to expand city capacity to identify and resolve housing-related public health problems This project uses machine learning to predict which properties in the city of Chel...
E-C113A90389	models/Predicting high risk violations.ipynb	models/Predicting high risk violations.ipynb	OBSERVED	[CODE FOR OUTPUT] chelsea = chelsea_df.copy() chelsea.shape [PERSISTED OUTPUT] (1611, 109) [CODE FOR OUTPUT] chelsea.head(2) [PERSISTED OUTPUT] Building Permits - Count See Click Fix - Co...
E-HIS-25C95144A8	git:063788fb91cdf160293b5ddf f07022ff8eafe4c1	git:063788fb91cdf160293b5ddf f07022ff8eafe4c1	OBSERVED	Commit: 063788fb91cdf160293b5ddf07022ff8eafe4c1 Authored: 2020-07-13T09:56:17-04:00 Subject: Add files via upload Body: Not recorded Changed files (3): A models/Predicting any code viola...
E-HIS-BB6B6668B7	git:de1f2e2e1b53e18805ba549 ba61214660d70d131	git:de1f2e2e1b53e18805ba549 ba61214660d70d131	OBSERVED	Commit: de1f2e2e1b53e18805ba549ba61214660d70d131 Authored: 2020-06-16T10:00:38-04:00 Subject: added data cleaning notebook Body: Not recorded Changed files (1): A data-cleaning/data-clean...
E-SYSTEM-PATH	.	.	OBSERVED	Imported directory name: R1_public-health_housing-chelsea

The PDF index includes evidence linked to material report content. 23 additional supporting History entries remain indexed and expandable in the complete HTML report.

Raw evidence excerpts

Raw excerpts are separated from the client narrative. Browser readers may expand every retained excerpt, and longer material is clearly marked as partial in print. The PDF includes bounded excerpts for evidence linked to material report content; supporting source-control History remains complete in the self-contained HTML report.

E-06B8ACEBB2 · data-cleaning/data-cleaning.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record. Source collection bounded this excerpt; the source may contain additional content. The HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
[CODE FOR OUTPUT]
print(chelsea.shape)
print(chelsea.columns)
```

[PERSISTED OUTPUT]

(7036, 68)

```
Index(['Parcel ID', 'Address',
      'Building Permits - Count - Between 01-01-2016 and 05-31-2019',
      'See Click Fix - Count - Between 08-01-2016 and 09-30-2019',
      'Certificate of Habitability Violations - Bathroom Lighting and Electrical Outlets - Between 08-01-2015 and 08-31-2019',
      'Certificate of Habitability Violations - Collection of Garbage and Rubbish - Between 08-01-2015 and 08-31-2019',
      ],
      dtype=object)
```

[Middle of retained excerpt omitted from the print edition.]

ents - Count	True	
Rat Baiting - Count		True
Tax Title		True
Total Assessed Val		

E-1352155D41 · models/Predicting any code violation.ipynb

Epistemic status: OBSERVED · Kind: notebook

This is the complete excerpt retained in the Project Record. Source collection bounded this excerpt; the source may contain additional content. The HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

[CODE FOR OUTPUT]
chelsea = chelsea_df.copy()
chelsea.shape

[PERSISTED OUTPUT]
(1611, 109)

[CODE FOR OUTPUT]
chelsea.head(2)

[PERSISTED OUTPUT]

	Building Permits - Count	See Click Fix - Count	Fire Incidents - Count	\
0	0.0	0.0	0.0	
1	2.0	0.0	0.0	

	Foreclosure Sales - Count	Vacant Foreclosed Properties	\
0	0.0	0.0	
1			

[Middle of retained excerpt omitted from the print edition.]

0.778151				
3	0.0	4.092229	5.844726	0.000000
4	0.0	3.715753	5.642761	0.602060

log_bathrooms	APT 4-UN
---------------	----------

E-5CD21A80C3 · models/Predicting overcrowding.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record. Source collection bounded this excerpt; the source may contain additional content. The HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
[CODE FOR OUTPUT]
chelsea = chelsea_df.copy()
chelsea.shape
```

```
[PERSISTED OUTPUT]
(1611, 109)
```

```
[CODE FOR OUTPUT]
chelsea.head(2)
```

```
[PERSISTED OUTPUT]
Building Permits - Count  See Click Fix - Count  Fire Incidents - Count  \
0                        0.0                    0.0                0.0
1                        2.0                    0.0                0.0
```

```
Foreclosure Sales - Count  Vacant Foreclosed Properties  \
0                        0.0                    0.0
1
```

[Middle of retained excerpt omitted from the print edition.]

```
1  \
0                        0                    0
1                        0                    0
2                        0                    0
3
```

E-B335630551 · README.md

Epistemic status: OBSERVED · **Kind:** documentation

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

Leveraging data to expand city capacity to identify and resolve housing-related public health problems

This project uses machine learning to predict which properties in the city of Chelsea, MA, will be exposed to public health hazards. The model works by training on a subsection of the city that was subject to a proactive code inspections program and then making predictions for the rest of the city.

Initial evaluation of the models suggests that the city government may use these results to be more effective at

[Middle of retained excerpt omitted from the print edition.]

rk was conducted as part of the Harvard Kennedy School's Ash Center for Democratic Governance and Innovation's [Innovation Field Lab](<https://ash.harvard.edu/innovation-field-lab>).

E-C113A90389 · models/Predicting high risk violations.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record. Source collection bounded this excerpt; the source may contain additional content. The HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

[CODE FOR OUTPUT]

```
chelsea = chelsea_df.copy()
chelsea.shape
```

[PERSISTED OUTPUT]

```
(1611, 109)
```

[CODE FOR OUTPUT]

```
chelsea.head(2)
```

[PERSISTED OUTPUT]

```
Building Permits - Count  See Click Fix - Count  Fire Incidents - Count  \
0                        0.0                      0.0                0.0
1                        2.0                      0.0                0.0
```

```
Foreclosure Sales - Count  Vacant Foreclosed Properties  \
0                        0.0                          0.0
1
```

[Middle of retained excerpt omitted from the print edition.]

```
nt(gridF.best_params_)
print(gridF.best_score_)
```

[PERSISTED OUTPUT]

```
{'criterion': 'gini', 'max_depth': 15, 'max_features': 'sqrt', 'min_samples_leaf': 1,
'min_samples_split': 2, 'n
```

E-HIS-25C95144A8 · git:063788fb91cdf160293b5ddff07022ff8eafe4c1

Epistemic status: OBSERVED · **Kind:** git commit

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Commit: 063788fb91cdf160293b5ddff07022ff8eafe4c1

Authored: 2020-07-13T09:56:17-04:00

Subject: Add files via upload

Body: Not recorded

Changed files (3):

```
A      models/Predicting any code violation.ipynb
A      models/Predicting high risk violations.ipynb
A      models/Predicting overcrowding.ipynb
```

E-HIS-BB6B6668B7 · git:de1f2e2e1b53e18805ba549ba61214660d70d131

Epistemic status: OBSERVED · **Kind:** git commit

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Commit: de1f2e2e1b53e18805ba549ba61214660d70d131
Authored: 2020-06-16T10:00:38-04:00
Subject: added data cleaning notebook
Body: Not recorded
Changed files (1):
A      data-cleaning/data-cleaning.ipynb
```

E-SYSTEM-PATH . .

Epistemic status: OBSERVED · **Kind:** project identity

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Imported directory name: R1_public-health_housing-chelsea
```

Bóveda processing metadata

Technical audit diagnostics are retained as a subordinate appendix and do not compete with the project narrative.

TELEMETRY AVAILABILITY complete	AUDIT DURATION 63.0 s
RECONSTRUCTION RESULT accepted	ATTEMPTS / PROVIDER CALLS 2 / 2
TOTAL RECORDED TOKENS 45,020	PAID MODEL ACTIVITY DURING STORED AUDIT Yes
LOCALISED REPAIR accepted · record.results_evaluation.other_material_result, record.results_evaluation.establishes	

Attempt 1 · full reconstruction

rejected

Duration	52.6 s
Recorded tokens	35,324
Validation findings	- record.results_evaluation.other_material_result: numeric value 59 is not grounded in cited evidence - record.results_evaluation.other_material_result: numeric value 50 is not grounded in cited evidence - record.results_evaluation.establishes: numeric value 348 is not grounded in cited evidence
Failure stage	grounding · record.results_evaluation.other_material_result: numeric value 59 is not grounded in cited evidence · record.results_evaluation.other_material_result: numeric value 50 is not grounded in cited evidence · record.results_evaluation.establishes: numeric value 348 is not grounded in cited evidence
Retry	The rejected fields were supplied to the next attempt as corrections for a localised repair.

Attempt 2 · localized repair
accepted

Duration	10.3 s
Recorded tokens	9,696
Validation findings	None recorded